

AI-POWERED TOOL RENTAL PLATFORM

Multi-Agent System

Technical Architecture & Agent Documentation

Project Type

ITI Graduation Project

Document Version

V2.0 — 2026

Technology Stack

OpenAI + ASP.NET Core + Angular

Total Agents

4 Specialized AI Agents

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1. System Overview

This document describes the five AI agents powering the Tool Rental Platform. Each agent is a focused microservice with a defined input, output, recommended model, and integration role within the overall system.

The platform enables users to list, search, and rent physical tools. AI agents enhance the experience by automating classification, image understanding, knowledge retrieval, and visual search.

#	Agent	Purpose	Recommended Model
1	Classification Agent	Understands user intent & identifies required tool	GPT-4o-mini / GPT-4.1-mini
2	Image Analysis Agent	Detects tool type, condition, and visible damage	GPT-4o / GPT-4.1 Vision
3	RAG Knowledge Base Agent	Answers queries from curated knowledge base	GPT-4o-mini + Qdrant + Embeddings
4	Image-to-Search Agent	Finds similar nearby tools from an uploaded image	GPT-4o Vision + DB Search

1 Classification Agent

Purpose	Understands user input and identifies the required tool category.
Input	Free-text string (Arabic or English) from the user.
Recommended Model	GPT-4o-mini / GPT-4.1-mini
Why This Model	Fast, low-cost, excellent at text classification, bilingual (AR/EN).
Output Format	JSON — category, confidence score, reason
Trigger	Every time a user types a tool search or request.

Output JSON

```
{
  "category": "Drill",
  "confidence": 0.92,
  "reason": "User wants to make a hole in a wall"
}
```

Use Cases

- User types "عايز أفتح حيطه" → Category: Drill
- User types "I need something to cut metal" → Category: Angle Grinder
- User types "محتاج أقطع خشب" → Category: Wood Saw
- Feeds the result to search filters automatically

Notes

- Confidence threshold should be ≥ 0.75 before auto-applying the category.
- Below threshold: prompt user to select manually.
- Supports code-switching (mixed Arabic/English input).

2 Image Analysis Agent

Purpose	Analyzes a tool image to detect type, condition, and visible damage.
Input	Image file (JPEG/PNG) uploaded by the tool owner.
Recommended Model	GPT-4o / GPT-4.1 Vision
Why Vision Model	Text models cannot interpret images. Vision capability is required.
Output Format	JSON — toolType, condition, visibleDamage[], confidence
Integration	Called when owner uploads a photo during tool listing.

Output JSON

```
{
  "toolType": "Drill",
  "condition": "Good",
  "visibleDamage": ["scratches", "dust"],
  "confidence": 0.87
}
```

Use Cases

- Owner uploads a photo → System auto-fills tool type field
- Condition auto-set to 'Good' → Pricing Agent uses condition factor = 1.0
- Damage list shown to renter as part of the listing transparency
- Low confidence (<0.75) → Owner asked to confirm tool type manually

Notes

- Always allow the owner to override the auto-detected values.
- Combine with Classification Agent output for higher accuracy.
- Store the raw image + analysis result in DB for audit purposes.

3 RAG Knowledge Base Agent

Purpose	Answers user questions from a curated knowledge base.
Input	User question in Arabic or English.
Embedding Model	text-embedding-3-small (OpenAI)
Vector Database	Qdrant (recommended) or Chroma
LLM	GPT-4o-mini
Knowledge Base Includes	Tool usage guides, safety instructions, rental policies, maintenance tips
Output Format	Formatted answer + cited source from the knowledge base

Output JSON

```
// Example Question:
// "أستخدم الشنور ازاى بأمان؟"

{
  "answer": "When using an angle grinder, always wear eye protection and gloves. Ensure the guard is in place before operating...",
  "source": "Safety Guide – Angle Grinder, Section 3.2",
  "confidence": 0.91
}
```

Use Cases

- "How do I use an angle grinder safely?" → Safety guide answer + source
- "What is the rental cancellation policy?" → Policy document answer
- "What drill bit should I use for tiles?" → Recommendation from usage guide
- "When should I service a chainsaw?" → Maintenance tip returned

Notes

- Chunk documents into 300–500 token segments before embedding.
- Use cosine similarity with a threshold of 0.75 for retrieval.
- Include top 3 matching chunks as context in the LLM prompt.
- Fallback: if no relevant chunk found, return a friendly 'not found' message.

4 Image-to-Search Agent

Purpose	Finds similar available tools nearby from an uploaded image.
Input	Image file (JPEG/PNG) + optional user location.
Step 1	Analyze image with GPT-4o Vision → extract tool type + features
Step 2	Query SQL Server DB by: category, location, availability
Recommended Model	GPT-4o Vision for image step + standard DB query
Output Format	JSON — detectedTool, similarToolsNearby[]

Output JSON

```
{
  "detectedTool": "Angle Grinder",
  "similarToolsNearby": [
    {
      "name": "Bosch Angle Grinder",
      "distance": "2.5 km",
      "pricePerDay": 120
    },
    {
      "name": "Makita Angle Grinder",
      "distance": "4.1 km",
      "pricePerDay": 100
    }
  ]
}
```

Use Cases

- User photographs a tool they need → System finds it nearby without typing
- Renter uploads an image from Google → Gets matching listings instantly
- Useful for users who do not know the tool's exact name

Notes

- Combine detected tool type with Classification Agent categories for consistency.
- Use geolocation (lat/lng) stored in the tools table for distance calculation.

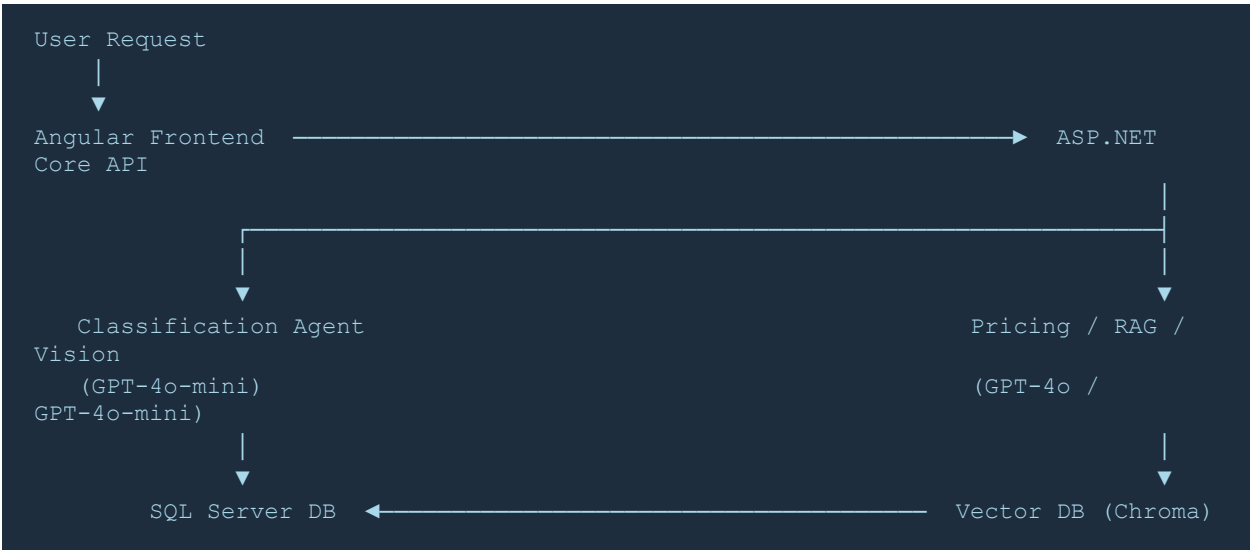
- Sort results by distance + price combined score.

7. Recommended System Architecture

The following stack is recommended for production deployment of the multi-agent tool rental platform:

Frontend	Angular — User interface, tool listing, image upload
Backend API	ASP.NET Core Web API — Business logic & agent orchestration
Database	SQL Server — Relational data (users, tools, rentals)
Vector Database	Chroma / MongoDB — Embeddings for RAG knowledge base
AI Provider	OpenAI API — GPT-4o, GPT-4o-mini, GPT-4.1 Vision
Authentication	ASP.NET Core Identity + JWT Tokens

Agent Data Flow



OpenAI Models — Quick Reference

Model	Speed	Cost	Best For
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GPT-4o-mini / GPT-4.1-mini	Very Fast	Very Low	Classification, Pricing, RAG
GPT-4o / GPT-4.1	Fast	Medium	Image Analysis, Vision tasks
text-embedding-3-small	Fast	Very Low	RAG Embeddings